

Sums of random variables, LLN and CLT

How the chances of a total are built, why an average settles down, and why totals end up bell-shaped.

THE QUESTION

What happens to the total and the average when we add up many independent random variables?

THE INTUITION

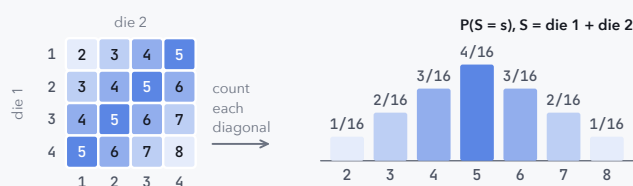
A total can be reached in many ways, so its PMF adds the chances of every pair that gives it: a convolution. Means add while SDs grow only like $\sqrt{n}$, so the average of n draws tightens around μ . Whatever one draw looks like, the total heads for a bell.

KEY TAKEAWAY

S_n has mean $n\mu$ and SD $\sigma\sqrt{n}$; $\bar{X}_n = S_n/n$ has mean μ and SD $\sigma/\sqrt{n}$. LLN: $\bar{X}_n \rightarrow \mu$. CLT: $\bar{X}_n$ is close to $\mathcal{N}(\mu, \sigma^2/n)$.

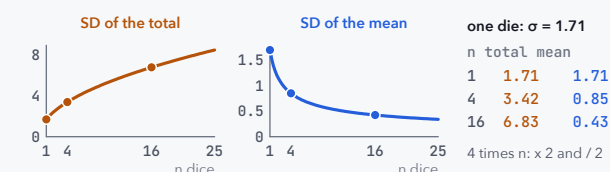
one draw $X \rightarrow$ add two: convolve $\rightarrow$ add n : SD $\sigma\sqrt{n} \rightarrow$ divide by n : SD $\sigma/\sqrt{n} \rightarrow$ LLN: mean $\rightarrow \mu \rightarrow$ CLT: bell shape

1 The PMF of a sum is a convolution



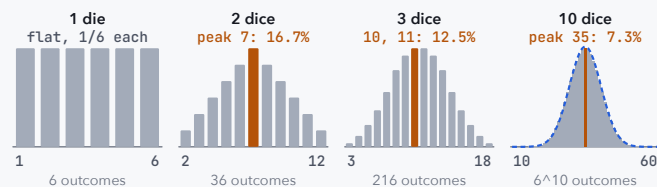
For independent X and Y , $P(X + Y = s) = \sum_x p_X(x)p_Y(s - x)$: walk along the diagonal $x + y = s$ and add the products. Two fair 4-sided dice have 16 equal pairs; the totals 2 to 8 take 1, 2, 3, 4, 3, 2, 1 of them, a triangle with its peak $1/4$ at 5. In NumPy, `np.convolve(p, q)`. A binomial is n Bernoullis convolved.

4 The mean of n draws has SD $\sigma/\sqrt{n}$



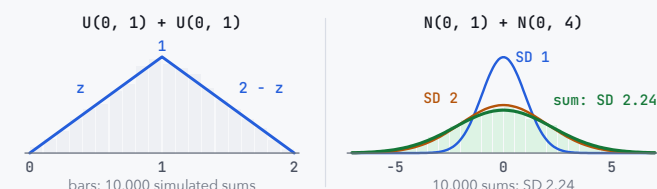
i.i.d. draws are independent with one distribution; their joint density is a product ($\mathcal{N}(0, 1)$ at 0.2 and 0.4: $0.391 \times 0.368 = 0.144$). With mean μ and variance σ^2 each, $E[S_n] = n\mu$ and $\text{Var}(S_n) = n\sigma^2$ (the expectation sheet). Divide by n : $E[\bar{X}_n] = \mu$, $\text{Var}(\bar{X}_n) = \sigma^2/n$. The SD $\sigma/\sqrt{n}$ is the standard error of the mean. Dependent draws add covariance terms (the covariance sheet).

2 Adding dice one at a time builds a bell



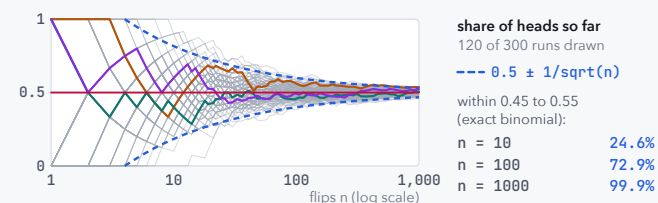
Each PMF is exact: convolve with one more die. One die is flat; two give the triangle with 7 at $6/36$; three are already rounded. Ten dice make a bell centred on 35 (7.27%) that sits on the normal curve with the same mean and SD (dashed). Nothing here started out normal; the extremes are rare because 10 and 60 each have one way out of 6^{10} .

3 Densities convolve too; normals stay normal



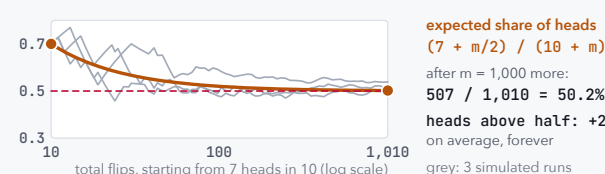
$f_{X+Y}(z) = \int f_X(x)f_Y(z - x) dx$. For two independent $U(0, 1)$ the overlap is z on $[0, 1]$ and $2 - z$ on $[1, 2]$: a triangle, with $P(0.5 < Z < 1.5) = 3/4$ (simulated 0.755). Independent normals add to a normal whose means and variances add: $\mathcal{N}(0, 1) + \mathcal{N}(0, 4) = \mathcal{N}(0, 5)$, SD $\sqrt{5} = 2.24$. Note: `torch.distributions.Normal(0, 2)` takes the SD 2, so it is $\mathcal{N}(0, 4)$ in mean-variance notation.

5 LLN: the running mean settles at μ



For i.i.d. draws with mean μ , $\bar{X}_n$ gets as close to μ as we like as n grows. Chebyshev on $\bar{X}_n$ shows it: $P(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2 / (n\varepsilon^2) \rightarrow 0$. For 100 fair flips and $\varepsilon = 0.1$ the bound is 0.25; the exact chance is 0.057. Each run wanders, but the band holding most runs narrows like $1/\sqrt{n}$.

6 Early luck gets diluted

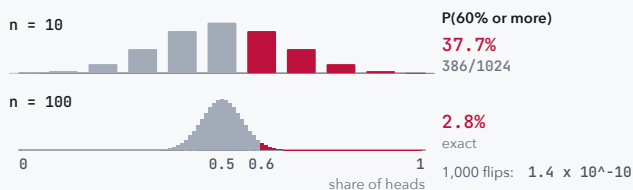


Seven heads in 10 flips, then 1,000 fair flips: expected heads $7 + 500 = 507$ of 1,010, a share of 50.2%. The coin has no memory, so the extra 2 heads stay on average; they just become a tiny share. Expecting tails "to even things out" is the gambler's fallacy. In the grey runs fresh luck (SD of the share about 0.016) swamps the early 2.

Sample means and the central limit theorem

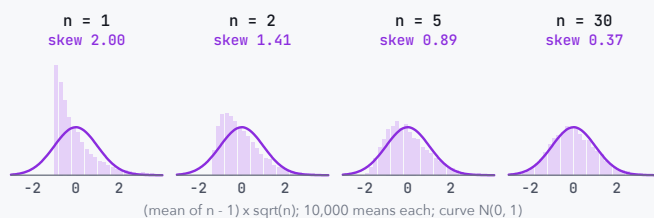
Small samples swing, the spread of a mean shrinks like $1/\sqrt{n}$, and the shape of a total turns normal.

7 Small samples swing more



The same fair coin gives 60% or more heads in 10 flips with chance $386/1024 = 37.7\%$, in 100 flips 2.8% and in 1,000 flips almost never. The share has SD $0.5/\sqrt{n}$: 0.158 for 10 flips, 0.05 for 100. So the highest and the lowest rates in a table often come from its smallest groups.

8 CLT: a standardised sum is close to $\mathcal{N}(0, 1)$

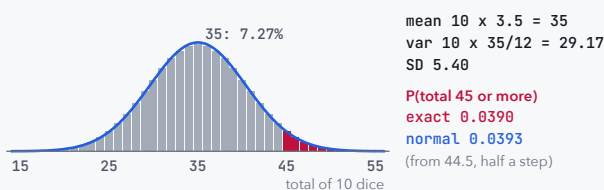


For i.i.d. X_i with mean μ and finite variance σ^2 ,

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

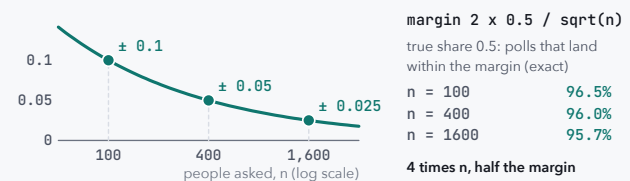
has a distribution that approaches $\mathcal{N}(0, 1)$, whatever one X_i looks like. So $S_n \approx \mathcal{N}(n\mu, n\sigma^2)$ and $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$. Exponential(1) waits have $\mu = \sigma = 1$ and skew 2; a mean of n has skew $2/\sqrt{n}$, so the bell arrives slowly. The LLN says where $\bar{X}_n$ goes; the CLT gives its shape around μ .

9 The normal curve approximates a dice total



Ten dice: mean 35, variance $10 \times 35/12 = 29.17$, SD 5.40. Exact $P(\text{total} \geq 45) = 0.0390$ by convolution; the normal from 44.5 gives $1 - \Phi(9.5/5.40) = 0.0393$. Starting at 45 gives 0.032: for whole-number totals, go half a step past the last value. $P(\text{total} \leq 30)$: exact 0.2050, normal 0.2024. At the peak, the curve's strip from 34.5 to 35.5 holds 0.0738 against the exact 0.0727.

10 Four times the sample halves the spread



The share $\hat{p}$ of n people who say yes has SD $\sqrt{p(1-p)/n} \leq 0.5/\sqrt{n}$, the standard error. By the CLT about 95% of such polls land within 2 SD of the true share, so the margin is about $2 \times 0.5/\sqrt{n}$: ± 0.10 at $n = 100$ and ± 0.05 at 400. It covers sampling luck only; asking more of the wrong people does not remove a bias.

11 Simulate both in a few lines

```
rng = np.random.default_rng(0)
K, N = 1000, 100 # runs, flips
x = rng.integers(0, 2, size=(K, N))
run = x.cumsum(1) / np.arange(1, N + 1)
m = run[:, -1] # final means
m.mean(), m.std() # 0.4996, 0.0500
(m >= 0.6).mean() # 0.028
```

```
p = np.ones(6) / 6 # one die
s = p
for _ in range(9): # 9 more dice
    s = np.convolve(s, p)
s.argmax() + 10, s.max() # 35, 0.0727
```

Each row of run is one LLN path; the 1,000 final means have SD $0.5/\sqrt{100} = 0.05$, and 2.8% of them reach 0.6, as the exact binomial says.

12 Pause and think

Two fair 4-sided dice: $P(\text{total} = 4)$? $3/16$: (1, 3), (2, 2), (3, 1).

The total of 3 dice: its variance? $3 \times 35/12 = 8.75$.

The mean of 36 dice rolls: its SD? $1.71/6 = 0.285$.

The share of heads in 400 fair flips: its SD?

$0.5/\sqrt{400} = 0.025$.

To halve the SD of a sample mean? Take 4 times as many draws.

520 heads in 1,000 flips. Is tails now due? No. The next flip is still 50-50.

Does the CLT make the data normal? No. Each draw keeps its shape; only the sum or mean of many turns bell-shaped.

PMF of a sum = convolution · $E[S_n] = n\mu$, $\text{SD}(S_n) = \sigma\sqrt{n}$ · $\text{SD}(\bar{X}_n) = \sigma/\sqrt{n}$ · LLN: $\bar{X}_n \rightarrow \mu$ · CLT: $\bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$

Source: the course notebooks on sums of random variables (two 4-sided dice; $\mathcal{N}(0, 1)$ plus a normal with SD 2 gives SD $\sqrt{5}$), the law of large numbers (running means of fair flips) and i.i.d. draws, and the LLN and CLT app, nipunbatra.github.io/psdv-teaching. Other examples and all numbers are our own, computed by the sheet's figure script (exact convolutions; simulations with seed 0). Next: the likelihood and log probabilities sheet.